MODEL LAB · 2026-10-15

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

26,804
held-out plate appearances

3.05%
held-out HR rate

0.6058
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-03, scored on the 26,804 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5778	1.66×	0.02953	0.13565	0.00621
KNN comps Prior is the pooled rate of the nearest comparables.	0.6058	1.55×	0.02947	0.13432	0.00421
SVR Support-vector regression on the same feature matrix.	0.5969	1.53×	0.02952	0.13512	0.00608
Random forest Forest regression; a learned similarity metric.	0.5978	1.53×	0.02951	0.13473	0.00580
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6058	1.55×	0.02947	0.13432	0.00421
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6058	1.55×	0.02947	0.13432	0.00421
Linear regression Ordinary least squares on the same features.	0.6003	1.53×	0.02951	0.13478	0.00444
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6008	1.64×	0.02953	0.13487	0.00658
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6001	1.58×	0.02950	0.13464	0.00483
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5977	1.55×	0.02954	0.13496	0.00695

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **257** knn **257** svr **257** rf **257** core **257** ensemble **257** linear **257** logistic **257** nn **257** xgb **257**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,077 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5874	1.25×	0.02287	0.11127	0.01213
KNN comps Prior is the pooled rate of the nearest comparables.	0.6175	1.94×	0.02281	0.10947	0.01009
SVR Support-vector regression on the same feature matrix.	0.5928	1.94×	0.02307	0.11387	0.01831
Random forest Forest regression; a learned similarity metric.	0.6057	2.36×	0.02302	0.11169	0.01872
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6175	1.94×	0.02281	0.10947	0.01009
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6175	1.94×	0.02281	0.10947	0.01009
Linear regression Ordinary least squares on the same features.	0.6136	1.94×	0.02308	0.11210	0.01708
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6205	1.94×	0.02312	0.11132	0.01760
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6203	2.22×	0.02286	0.10990	0.01274
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6098	1.94×	0.02329	0.11305	0.02431

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Shohei Ohtani 17.1%, Jackson Chourio 16.9%, Jonathan Aranda 15.4%, Jake Bauers 15.2%, William Contreras 15.2%, Junior Caminero 15.0%
Flat league	6/6	Jackson Chourio 17.0%, Shohei Ohtani 15.5%, William Contreras 15.0%, Jonathan Aranda 14.6%, Junior Caminero 14.5%, Jake Bauers 14.3%
KNN comps	6/6	Jackson Chourio 17.1%, Shohei Ohtani 17.0%, Jonathan Aranda 15.6%, Jake Bauers 15.5%, William Contreras 15.2%, Junior Caminero 14.9%
SVR	6/6	Jackson Chourio 17.6%, Shohei Ohtani 17.0%, Junior Caminero 16.2%, Jake Bauers 15.6%, Jonathan Aranda 15.0%, William Contreras 15.0%
Random forest	6/6	Jackson Chourio 17.7%, Shohei Ohtani 17.2%, Jake Bauers 15.7%, Junior Caminero 15.3%, Jonathan Aranda 15.2%, William Contreras 15.0%
Core (KNN+SVR)	6/6	Jackson Chourio 17.1%, Shohei Ohtani 17.0%, Jonathan Aranda 15.6%, Jake Bauers 15.5%, William Contreras 15.2%, Junior Caminero 14.9%
Forest blend	6/6	Jackson Chourio 17.1%, Shohei Ohtani 17.0%, Jonathan Aranda 15.6%, Jake Bauers 15.5%, William Contreras 15.2%, Junior Caminero 14.9%
Linear regression	5/6	Jackson Chourio 17.4%, Shohei Ohtani 17.0%, Jonathan Aranda 15.3%, Jake Bauers 15.3%, Will Smith 15.1%, William Contreras 15.1%
Logistic regression	6/6	Jackson Chourio 17.5%, Shohei Ohtani 17.4%, Junior Caminero 15.5%, Jake Bauers 15.3%, William Contreras 15.1%, Jonathan Aranda 15.1%
Neural net (torch)	6/6	Jackson Chourio 17.5%, Shohei Ohtani 17.3%, Junior Caminero 15.5%, Jake Bauers 15.4%, Jonathan Aranda 15.2%, William Contreras 15.1%
XGBoost	6/6	Jackson Chourio 18.0%, Shohei Ohtani 17.7%, Junior Caminero 15.8%, Jake Bauers 15.7%, Jonathan Aranda 15.4%, William Contreras 15.2%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TC
1 Shohei Ohtani slate	**17.1%**	15.5%	17.0%	17.0%	17.2%	17.0%	17.0%	17.0%	17.4%	
2 Jackson Chourio slate	**16.9%**	17.0%	17.1%	17.6%	17.7%	17.1%	17.1%	17.4%	17.5%	
3 Jonathan Aranda slate	**15.4%**	14.6%	15.6%	15.0%	15.2%	15.6%	15.6%	15.3%	15.1%	
4 Jake Bauers MIL	**15.2%**	14.3%	15.5%	15.6%	15.7%	15.5%	15.5%	15.3%	15.3%	
5 William Contreras MIL	**15.2%**	15.0%	15.2%	15.0%	15.0%	15.2%	15.2%	15.1%	15.1%	
6 Junior Caminero TB	**15.0%**	14.5%	14.9%	16.2%	15.3%	14.9%	14.9%	15.1%	15.5%	
7 Will Smith LAD	**14.5%**	14.2%	14.4%	14.4%	14.9%	14.4%	14.4%	15.1%	14.9%	
8 Teoscar Hernández LAD	**14.5%**	13.7%	14.7%	14.5%	14.7%	14.7%	14.7%	14.6%	14.7%	
9 Jo Adell CLE slate	**13.8%**	13.1%	13.5%	14.0%	13.6%	13.5%	13.5%	13.7%	14.1%	
10 Mookie Betts LAD	**13.8%**	13.7%	13.6%	14.0%	13.7%	13.6%	13.6%	13.8%	13.8%	
11 Victor Mesa Jr. TB	**13.8%**	13.6%	14.1%	14.6%	14.3%	14.1%	14.1%	14.2%	14.1%	
12 Brice Turang MIL	**13.5%**	13.8%	13.4%	13.3%	13.2%	13.4%	13.4%	12.8%	12.7%	
13 Patrick Bailey CLE	**13.4%**	12.7%	13.4%	13.2%	12.9%	13.4%	13.4%	13.2%	12.8%	
14 Freddie Freeman LAD	**13.3%**	12.9%	13.1%	12.9%	13.1%	13.1%	13.1%	13.2%	13.0%	
15 Christian Yelich MIL	**12.3%**	12.5%	12.6%	12.2%	12.6%	12.6%	12.6%	12.7%	12.2%	
16 Garrett Mitchell MIL	**12.3%**	11.7%	12.1%	11.0%	11.4%	12.1%	12.1%	11.5%	11.5%	
17 Chase DeLauter CLE	**12.3%**	12.3%	12.2%	12.0%	12.1%	12.2%	12.2%	12.1%	11.9%	
18 Yandy Díaz TB	**11.8%**	12.2%	11.9%	12.4%	12.2%	11.9%	11.9%	12.0%	11.9%	
19 Travis Bazzana CLE	**11.7%**	11.7%	11.5%	11.4%	11.5%	11.5%	11.5%	11.6%	11.3%	
20 Nathaniel Lowe CLE	**11.5%**	11.3%	11.7%	11.4%	11.7%	11.7%	11.7%	11.5%	11.4%	
21 Kyle Tucker LAD	**11.4%**	11.5%	11.4%	11.1%	11.4%	11.4%	11.4%	11.3%	11.0%	
22 Enrique Hernández LAD	**11.2%**	11.0%	11.2%	10.9%	11.2%	11.2%	11.2%	10.9%	10.6%	
23 Andy Pages LAD	**10.8%**	10.4%	10.5%	10.3%	10.5%	10.5%	10.5%	10.4%	10.2%	

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
24 **José Ramírez** CLE	**10.6%**	10.9%	10.7%	10.8%	10.8%	10.7%	10.7%	11.0%	11.2%	
25 **Petey Halpin** CLE	**8.5%**	9.6%	8.6%	9.0%	8.8%	8.6%	8.6%	9.6%	9.0%	
26 **Miguel Rojas** LAD	**7.9%**	9.7%	8.2%	8.3%	8.0%	8.2%	8.2%	8.6%	8.7%	
27 **Liam Hicks** TB	**7.3%**	8.4%	7.2%	8.1%	7.3%	7.2%	7.2%	7.7%	7.5%	
28 **Brayan Rocchio** CLE	**7.3%**	7.8%	7.2%	7.3%	6.9%	7.2%	7.2%	7.0%	7.0%	
29 **Jonny DeLuca** TB	**7.2%**	7.8%	7.2%	7.1%	6.7%	7.2%	7.2%	7.0%	7.3%	
30 **Richie Palacios** TB	**7.0%**	8.0%	7.1%	6.8%	6.5%	7.1%	7.1%	6.9%	6.9%	
31 **Cooper Pratt** MIL	**6.6%**	8.5%	6.6%	5.6%	6.2%	6.6%	6.6%	5.7%	6.0%	
32 **Sal Frelick** MIL	**6.5%**	8.2%	6.7%	6.4%	6.3%	6.7%	6.7%	6.4%	6.5%	
33 **David Hamilton** MIL	**6.1%**	7.7%	6.0%	5.9%	5.7%	6.0%	6.0%	6.1%	6.0%	
34 **Steven Kwan** CLE	**5.3%**	8.7%	5.7%	3.7%	4.3%	5.7%	5.7%	3.7%	5.3%	
35 **Taylor Walls** TB	**4.8%**	6.6%	4.9%	4.1%	4.1%	4.9%	4.9%	5.0%	5.0%	
36 **Chandler Simpson** TB	**2.9%**	6.3%	3.7%	2.3%	1.7%	3.7%	3.7%	1.7%	3.3%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-15 · model 2026.10.01-stack · built 2026-10-15 22:15:22 · live data 2026-10-15 22:15:22

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